Programmin with Concurrent Objects

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Introduction

The Hungry Birds Problem

Implementation

We created multiple threads for the baby birds and the parent bird. Each of the baby birds constantly tries to eat worms from the plate. The first baby bird that finds the plate empty signals the parent by chirping. We used a boolean variable "signal" that is used to ensure only the first bird that finds the plate will notify the parent. All other baby birds wait until the plate is refilled. Once the baby bird signals the parent, it refills the plate and notifies all the baby birds so they can continue to eat.

Results

When we run the code, the baby birds continuously eat from the plate until it is empty and one baby bird announces the empty plate by chirping, the parent bird refills the plate with 10 new worms, announces that the plate was refilled and the baby birds continue eating.

The Bear and Honeybees Problem

Implementation

We created multiple threads for the bees and one for the bear. Each of the bees constantly tries to fill the honeypot. The bee that fills the honeypot will notify all other bees and the bear. All bees will wait until the honeypot is emptied again by the bear, and start filling it up again. The bear can only empty the honeypot once full, ensuring fairness.

Results

When we run the code, the bees continuously fill from the honeypot until it is full and the bee that filled the honeypot announces the full honeypot, waking all waiting threads up. The bear empties the honeypot and announces to all the bees that the honeypot was emptied. the bees continue filling it up again.